Zhiying Wang

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EDUCATION

North Carolina State University Raleigh, NC, USA
Department of Computer Science August 2025 – Present
PhD in Computer Science
Research Interests: Satellite resource scheduling, edge computing, and optimization methods.

University of Electronic Science and Technology of China Chengdu, Sichuan, China
School of Information and Communication Engineering September 2022 – June 2025
MS in Electronic Information Engineering, GPA: 3.68/4.0
Relevant Coursework: Basis of communication network system; Optimization Methods and Applications; Information Theory; Graph Theory and its Application.

Hebei University of Technology Tianjin, China
School of Electronic Information Engineering September 2018 – June 2022
BE in Electronic Information Engineering, GPA: 3.87/4.0
Relevant Coursework: Advanced Mathematics, Linear Algebra, Probability and Statistics, Digital Signal Processing, Signal and Linear System, Data Communication and Computer Network, Theory of Communication.

PUBLICATIONS

- **Zhiying Wang**, Xiaojian Wang, Huayue Gu, Zhishan Guo, and Ruozhou Yu. "STAR-GS: Truthful and Visibility-Aware Resource Scheduling for Ground Station as a Service," in *IEEE International Conference on Network Protocols (ICNP)*, 2026.
- **Zhiying Wang**, Jinghui Chen, Gang Sun, Hongfang Yu, Mohsen Guizani. "Joint Energy-Efficient Task Scheduling and Trajectory Optimization for Multi-UAV MEC Networks in Low-Altitude Economy," in *IEEE Transactions on Cognitive Communications and Networking (TCCN)*, vol. 12, pp. 3058–3072, 2025.
- **Zhiying Wang**, Tianxi Wei, Gang Sun, Xinyue Liu, Hongfang Yu, Dusit Niyato. "Multi-UAV Enabled MEC Networks: Optimizing Delay through Intelligent 3D Trajectory Planning and Resource Allocation," in *IEEE Transactions on Intelligent Transportation Systems (TITS)*, vol. 26, pp. 20897–20911, 2025.
- **Zhiying Wang**, Gang Sun, Yuhui Wang, Hongfang Yu, Dusit Niyato. "Cluster-Based Multi-Agent Task Scheduling for Space-Air-Ground Integrated Networks," in *IEEE Transactions on Cognitive Communications and Networking (TCCN)*, vol. 12, pp. 29–42, 2025.
- **Zhiying Wang**, Gang Sun, Hanyue Su, Hongfang Yu, Bo Lei, and Mohsen Guizani. "Low-Latency Scheduling Approach for Dependent Tasks in MEC-Enabled 5G Vehicular Networks," in *IEEE Internet of Things Journal (IoTJ)*, vol. 11, no. 4, pp. 6278–6289, Feb. 15, 2024.
- Gang Sun, **Zhiying Wang**, Hanyue Su, Hongfang Yu, Bo Lei, and Mohsen Guizani. "Profit Maximization of Independent Task Offloading in MEC-Enabled 5G Internet of Vehicles," in *IEEE Transactions on Intelligent Transportation Systems (TITS)*, pp. 1–13, 2024.
- **Zhiying Wang**, Guanhua Huang, Gang Sun, Hongfang Yu, Jian Sun. "Reinforcement Q-Learning Enabled Energy-Efficient Service Function Chain Provisioning in Multi-Domain Networks," in *Peer-to-Peer Networking and Applications*, pp. 1–19, 2024.

RESEARCH INTERESTS AND DIRECTIONS

- **Mobile Edge Computing (MEC)**: Developing algorithms and systems that leverage the proximity of edge computing resources to reduce latency, enhance data processing speeds, and improve the overall efficiency of mobile networks.
- **Deep Reinforcement Learning (DRL)**: Crafting innovative deep learning models that can learn and adapt through interaction with their environment, aiming to solve complex decision-making and optimization problems.
- **Internet of Things (IoT)**: Designing and implementing IoT systems that intelligently connect and manage a vast network of devices, enabling smarter environments and enhancing human-technology interactions.
- **Unmanned Aerial Vehicles (UAV)**: Investigating the use of UAVs in various applications, from surveillance and delivery to disaster management, focusing on autonomous flight, navigation, and communication technologies.

PROJECT EXPERIENCES

Optimization Algorithms in UAV-Assisted Mobile Edge Computing Networks July 2022 - today
University of Electronic Science and Technology of China
Role: Leader of my graduation project

- Multi-UAV Edge Computing Network Simulation: Developed and simulated models for UAV-assisted edge computing, establishing a base for performance optimization.
- DRL-Based UAV Trajectory Optimization: Enhanced UAV operational efficiency by optimizing flight trajectories using the DRL algorithm.

- Game Theory for Efficient Task Scheduling: Optimized UAV task distribution using game theory, ensuring optimal network performance and task allocation.

Key Information of Internet Infrastructure Services Platform (KI3) October 2022 - March 2023
Led by: Tsinghua University
Role: Representative from the University of Electronic Science and Technology of China, Collaborator

- Satellite Communications Research: Led in-depth research on satellite communication technologies with a focus on operational insights and technical specifications.
- Starlink Data Management: Spearheaded the collection and systematic storage of Starlink satellite and ground station data for analytical use.
- Simulation and Modeling Assistance: Enabled precise satellite and ground station modeling simulations for engineers by utilizing geographic and TLE data insights.

American Mathematical Contest in Modeling (MCM) February 2021
Hebei University of Technology
Role: Leader

- Modeling: Developed a PCA-based fungal interaction model, analyzing key biological metrics.
- Leadership and Learning: Led team and self-learned mathematical modeling and LaTeX typesetting.
- Achievement: Secured the Finalist Prize(top 2.8%) with exceptional preparation and effort.

FELLOWSHIPS & AWARDS

- Graduate National Scholarship, 2024
- Sichuan Province Outstanding Graduate, 2024
- University of Electronic Science and Technology of China, First-Class Scholarship, 2024
- University of Electronic Science and Technology of China, First-Class Scholarship, 2023
- University of Electronic Science and Technology of China, First-Class Scholarship, 2022
- Hebei University of Technology, First-Class Scholarship, 2021
- American Mathematical Contest in Modeling (MCM), Finalist(top 2.8%), 2021
- Hebei University of Technology, Second-Class Scholarship, 2020
- Hebei University of Technology, First-Class Scholarship, 2019
- National College Student English Competition, Third Prize, 2019

Academic Service

Reviewer

- *IEEE Internet of Things Journal (IoTJ)*
- *IEEE Transactions on Mobile Computing (TMC)*
- *IEEE GLOBECOM Workshops*
- *International Wireless Communications & Mobile Computing Conference (IWCMC)*
- *Euromicro Conference on Digital System Design (DSD)*
- *IEEE Network*

Conference and Workshop Roles

- Track Chair, *DLCV Workshop*, International Conference on Deep Learning and Vision (ICDL-V), 2026

Teaching Experience

Teaching Assistant, North Carolina State University Raleigh, NC, USA

- **CSC 570: Computer Networks**, Spring 2026
  - Organized and facilitated student Q&A sessions to reinforce key networking concepts
  - Provided individualized support on course material, assignments, and technical questions
  - Assisted with grading and provided constructive feedback on student work
- **CSC 316: Data Structures and Algorithms**, Fall 2025
  - Held weekly office hours to explain course concepts and provide individualized academic support
  - Graded programming assignments and written homework
  - Proctored examinations and assisted with classroom logistics

SKILLS & INTERESTS

Skills: Python; Pytorch; Latex; MATLAB; STK(Satellite Tool Kit); Mininet

- Programming & Modeling: Proficient in Python for modeling and algorithm development; experienced with MATLAB, STK and Mininet for satellite and network simulations.
- Deep Learning & Optimization: Skilled in Pytorch for Deep Reinforcement Learning model training; knowledgeable in convex optimization and graph theory for problem-solving.
- Technical Writing: Proficient in Latex for high-quality academic and technical documentation.

Language: English; Chinese
Interests: Tennis; Cycling